

COMP1011 Assignment 2

- **Assignment Deadline: 24 April 2025, 11:59PM**
- **Submission: 1) a cpp file with student ID and name, 2) a screenshot of code and result**

Problem: Doubly Linked List

Develop the following four functions for doubly linked lists (i.e., each node contains the data, a link to the previous node, and a link to the next node):

- **void printList(Node *head);**

// It prints all nodes of the linked list one by one.
- **void insertList(Node **headptr, int value);**

// It inserts a new node to the linked list such that all nodes are still sorted in ascending order after the insertion.
- **bool isSortedList(Node *head);**

// It checks whether all nodes of the linked list are sorted in ascending order. It returns true if the nodes are sorted; otherwise false.

// An empty linked list is regarded as sorted.
- **Node* mergeSortedList(Node *head1, Node *head2)**

// It merges two sorted linked lists into a single sorted linked list and returns the head of the merged list.

// It should be able to handle empty linked lists.

// No new node should be created in this function.

Please find a template cpp file on Blackboard, with the main() function to test your code.